

Ans:

(1)

$$a) U = (xy)^2$$

Multiplying U by θ ,

$$(\theta x \cdot \theta y)^2$$

$$= (\theta^2 xy)^2$$

$$= \theta^4 (xy)^2$$

$$= \theta^4 U$$

Thus this utility function is homogeneous of degree 4.

$$b) U = (xy)^{1/2}$$

Multiplying U by θ ,

$$(\theta x \cdot \theta y)^{1/2}$$

$$= (\theta^2 xy)^{1/2}$$

$$= \theta \cdot (xy)^{1/2}$$

$$= \theta \cdot U$$

Thus this utility function is homogeneous of degree 1.

$$c) U = \min(x, 2y)$$

Multiplying U by θ ,

$$\begin{aligned} & \min(\theta x, \theta 2y) \\ &= \theta \min(x, 2y) \\ &= \theta \cdot U \end{aligned}$$

Thus, homogeneous of degree of this utility function is 1.

$$d) U = 2x + y$$

Multiplying U by θ ,

$$\begin{aligned} & \theta 2x + \theta y \\ &= \theta(2x + y) \\ &= \theta \cdot U \end{aligned}$$

Thus this utility function is homogeneous of degree 1.

$$e) U = x^r y^r + xy$$

Multiplying U by θ ,

$$\begin{aligned} & (\theta x)^r (\theta y)^r + (\theta x)(\theta y) \\ &= \theta^r x^r \cdot \theta^r y^r + \theta^r xy \\ &= \theta^4 (x^r y^r) + \theta^r (xy) \end{aligned}$$

This function is not homogeneous.

Ans:

②

A function is homothetic if it is a monotonic transformation of a homogeneous function.

From ①, we found the ~~homogeneity~~ homogeneous degree of all functions, so —

① $U = (xy)^2$; This function is homogeneous and so is homothetic.

② $U = (xy)^{1/2}$; This function is homogeneous degree of 1 and homothetic.

③ $U = \min(x, 2y)$; This function is homogeneous degree of 1 and ~~the~~ homothetic.

④ $U = 2x + y$; This function is homogeneous degree of 1 and so is homothetic.

⑤ $U = x^2y + xy$; This function is not homogeneous. Since homothetic functions are typically derived from homogeneous functions, this function is not homothetic either.

Need to show MRS_{xy}

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$$MRS_{xy} = \frac{\partial U / \partial x}{\partial U / \partial y} = \frac{y}{x}$$

$\Rightarrow h.d. 0$

Ans:

(3)

(a) $U = (xy)^r$

Here, $\max_{x,y} U = (xy)^r = x^r y^r$
s.t. $I = P_x x + P_y y$

Thus, the Lagrangian:

$$L = x^r y^r + \lambda (I - P_x x - P_y y)$$

FONC requires,

$$\frac{\partial L}{\partial x} = 2x^{r-1} y^r - \lambda P_x = 0 \quad \text{or, } 2x^{r-1} y^r = \lambda P_x \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial y} = 2x^r y^{r-1} - \lambda P_y = 0 \quad \text{or, } 2x^r y^{r-1} = \lambda P_y \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = I - P_x x - P_y y = 0 \quad \text{or, } I = P_x x + P_y y \quad \text{--- (iii)}$$

Now, dividing (i) by (ii),

$$\frac{2x^{r-1} y^r}{2x^r y^{r-1}} = \frac{\lambda P_x}{\lambda P_y} \Rightarrow \frac{y}{x} = \frac{P_x}{P_y} \Rightarrow y = \frac{P_x x}{P_y} \quad \text{--- (iv)}$$

Putting value of y in (iii);

$$I = P_x x + P_y \frac{P_x x}{P_y}$$

$$\Rightarrow I = P_x x + P_x x$$

$$\Rightarrow I = 2P_x x$$

$$\therefore x^* = \frac{I}{2P_x}$$

Putting value of x^* in (iv),

$$y^* = \frac{P_x}{P_y} \cdot \frac{I}{2P_x} = \frac{I}{2P_y}$$

So, the Marshallian demands are —

$$x^* = \frac{I}{2P_x} \quad \text{and} \quad y^* = \frac{I}{2P_y}$$

Now, Indirect utility function —

$$\begin{aligned} V(P_x, P_y, I) &= \left(\frac{I}{2P_x} \cdot \frac{I}{2P_y} \right)^{\frac{1}{2}} \\ &= \left(\frac{I^2}{4P_x P_y} \right)^{\frac{1}{2}} \\ &= \frac{I^1}{16 P_x^{\frac{1}{2}} P_y^{\frac{1}{2}}} \end{aligned}$$

∴ The indirect utility function is $\frac{I^1}{16 P_x^{\frac{1}{2}} P_y^{\frac{1}{2}}}$.

$$(b) U = (xy)^{1/2}$$

$$\text{Here, } \max_{x,y} U = (xy)^{1/2}$$

$$\text{s.t. } I = P_x x + P_y y$$

The Lagrangian,

$$L = (xy)^{1/2} + \lambda(I - P_x x - P_y y)$$

FONC requires,

$$\frac{\partial L}{\partial x} = \frac{1}{2} x^{-1/2} y^{1/2} - \lambda P_x = 0 \Rightarrow \frac{y^{1/2}}{2 x^{1/2}} = \lambda P_x \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial y} = \frac{1}{2} x^{1/2} y^{-1/2} - \lambda P_y = 0 \Rightarrow \frac{x^{1/2}}{2 y^{1/2}} = \lambda P_y \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = I - P_x x - P_y y = 0 \Rightarrow I = P_x x + P_y y \quad \text{--- (iii)}$$

Dividing (i) by (ii) -

$$\frac{y^{1/2} / 2 x^{1/2}}{x^{1/2} / 2 y^{1/2}} = \frac{\lambda P_x}{\lambda P_y}$$

$$\Rightarrow \frac{y^{1/2+1/2}}{x^{1/2+1/2}} = \frac{P_x}{P_y}$$

$$\Rightarrow \frac{y}{x} = \frac{P_x}{P_y}$$

$$\therefore y = \frac{P_x x}{P_y} \quad \text{--- (iv)}$$

Putting value of y in (iii).

$$I = P_x x + P_y \frac{P_x x}{P_y}$$

$$\Rightarrow I = P_x x + P_x x$$

$$\Rightarrow x^* = \frac{I}{2P_x}$$

Putting the value of x^* in (iv),

$$y^* = \frac{P_x}{P_y} \cdot \frac{I}{2P_x}$$

$$= \frac{I}{2P_y}$$

Thus, Marshallian demands are —

$$x^* = \frac{I}{2P_x} \text{ and } y^* = \frac{I}{2P_y}$$

Indirect utility function —

$$v(P_x, P_y, I) = \left(\frac{I}{2P_x} \cdot \frac{I}{2P_y} \right)^{1/2}$$

$$= \left(\frac{I^2}{4P_x P_y} \right)^{1/2}$$

$$= \frac{I}{2\sqrt{P_x P_y}}$$

The indirect utility function is $\frac{I}{2\sqrt{P_x P_y}}$.

(c) $U = \min(x, 2y)$

$x = 2y$ since utility is maximized when $x = 2y$.

The budget constraint —

$$I = P_x x + P_y y$$

$$\Rightarrow I = P_x \cdot 2y + P_y y$$

$$\Rightarrow I = y(2P_x + P_y)$$

$$\therefore y^* = \frac{I}{2P_x + P_y}$$

$$\therefore x^* = \frac{2I}{2P_x + P_y}$$

Marshallian demands are —

$$x^* = \frac{2I}{2P_x + P_y} \quad \text{and} \quad y^* = \frac{I}{2P_x + P_y}$$

Indirect utility function —

$$\begin{aligned} V(P_x, P_y, I) &= \min \left(\frac{2I}{2P_x + P_y}, \frac{2I}{2P_x + P_y} \right) \\ &= \frac{I}{2P_x + P_y} \end{aligned}$$

d) ~~Utility~~ $U = 2x + y$

$$U = 2x + y$$

This utility function is ~~linear~~ linear.
Here

The marginal utility of x is 2, so the marginal utility per dollar for x is $\frac{2}{P_x}$.

The marginal utility of y is 1, so the marginal utility per dollar for y is $\frac{1}{P_y}$.

Now,

If, $\frac{2}{P_x} > \frac{1}{P_y}$, the consumer will spend all income on x ,

$$\text{so, } x = \frac{I}{P_x}, y = 0$$

If $\frac{2}{P_x} < \frac{1}{P_y}$, the consumer will spend all income on y ,

$$\text{so, } x = 0, y = \frac{I}{P_y}$$

If $\frac{2}{P_x} = \frac{1}{P_y}$, the consumer is indifferent and can choose any combination along the budget line.

The indirect utility function,

$$V(P_x, P_y, I) = \begin{cases} \frac{2I}{P_x} & \text{if } \frac{2}{P_x} > \frac{1}{P_y} \\ \frac{I}{P_y} & \text{if } \frac{2}{P_x} < \frac{1}{P_y} \end{cases}$$

$$c) U = x^r y^r + xy$$

$$\cdot \text{Max}_{x,y} U = x^r y^r + xy$$

$$\text{s.t } I = P_x x + P_y y$$

The Lagrangian,

$$L = x^r y^r + xy - \lambda(I - P_x x - P_y y)$$

Here, FONC -

$$\frac{\partial L}{\partial x} = 2x^{r-1}y^r + y - \lambda P_x = 0$$

$$\Rightarrow 2x^{r-1}y^r + y = \lambda P_x \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial y} = 2x^r y^{r-1} + x - \lambda P_y = 0$$

$$\Rightarrow 2x^r y^{r-1} + x = \lambda P_y \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = I - P_x x - P_y y = 0$$

$$\Rightarrow I = P_x x + P_y y \quad \text{--- (iii)}$$

Dividing by (i) and (ii)

$$\frac{2x^{r-1}y^r + y}{2x^r y^{r-1} + x} = \frac{\lambda P_x}{\lambda P_y}$$

$$\Rightarrow \frac{y(2x^{r-1}y^r + 1)}{x(2x^r y^{r-1} + 1)} = \frac{P_x}{P_y}$$

$$\Rightarrow \frac{y}{x} = \frac{P_x}{P_y}$$

$$\therefore y = \frac{P_x x}{P_y} \quad \text{--- (iv)}$$

Putting value of y in (iii),

$$I = P_x x + P_y \cdot \frac{P_x x}{P_y}$$

$$\Rightarrow I = P_x x + P_x x$$

$$\Rightarrow I = 2P_x x$$

$$\Rightarrow x^* = \frac{I}{2P_x}$$

Putting the value x in (iv) $\Rightarrow$

$$y^* = \frac{I}{2P_y}$$

The Marshallian demands are—

$$x^* = \frac{I}{2P_x} \text{ and } y^* = \frac{I}{2P_y}$$

The indirect utility function,

$$V(P_x, P_y, I) = \left(\frac{I}{2P_x}\right)^{\alpha} \left(\frac{I}{2P_y}\right)^{\beta} + \frac{I}{2P_x} \cdot \frac{I}{2P_y}$$

$$= \frac{I^{\alpha+\beta}}{16 P_x^{\alpha} P_y^{\beta}} + \frac{I^2}{4P_x P_y}$$

$$= \frac{I^2}{4P_x P_y} \left(\frac{I^{\alpha+\beta-2}}{4P_x^{\alpha} P_y^{\beta}} + 1 \right).$$

Ans

(4)

$$a) U = (xy)^r = x^r y^r$$

Marginal utilities are —

$$MU_x = \frac{\partial U}{\partial x} = 2xy^r \cdot y = 2xy^3$$

$$MU_y = \frac{\partial U}{\partial y} = 2x^r y \cdot x = 2x^3 y$$

Since each marginal utility depends on both x and y , increasing x or y leads to a non-linear increase in the marginal utility, suggesting that this utility function does not exhibit diminishing marginal utility. Instead, it has increasing marginal utility due to the squared terms.

$$b) U = (xy)^{1/2}$$

Marginal Utilities are —

$$MU_x = \frac{\partial U}{\partial x} = \frac{1}{2}(xy)^{-1/2} \cdot y = \frac{y}{2\sqrt{xy}}$$

$$MU_y = \frac{\partial U}{\partial y} = \frac{1}{2}(xy)^{-1/2} \cdot x = \frac{x}{2\sqrt{xy}}$$

Hence, as x or y increases, the marginal utility of each good decreases due to the $(xy)^{-1/2}$ factor. Therefore, this function does exhibit diminishing marginal utility.

d) $U = 2x + y$

Marginal Utilities are —

$$MU_x = \frac{\partial U}{\partial x} = 2$$

$$\frac{\partial^2 U}{\partial x^2} = 0$$

$$MU_y = \frac{\partial U}{\partial y} = 1$$

$$\frac{\partial^2 U}{\partial y^2} = 0$$

For this linear utility function, the marginal utility of each good is constant. Thus, it does not exhibit diminishing marginal utility.

e) $U = x^2y + xy$

Marginal utilities are —

$$MU_x = \frac{\partial U}{\partial x} = 2xy + y$$

$$MU_y = \frac{\partial U}{\partial y} = x^2 + x$$

For this function, marginal utility does not necessarily diminish, it ~~does~~ may vary irregularly due to the nonlinear terms. So, it does not consistently exhibit diminishing marginal utility.

Ans.

(5)

$$\textcircled{b} U = (xy)^{1/2}$$

$$\text{Min}_{x,y} P_x x + P_y y$$

$$\text{s.t. } (xy)^{1/2} = \bar{U} \Rightarrow xy = \bar{U}^2$$

The Lagrangian,

$$L = P_x x + P_y y + \lambda (\bar{U}^2 - xy)$$

FONC requires,

$$\frac{\partial L}{\partial x} = P_x - \lambda y = 0 \Rightarrow \lambda = \frac{P_x}{y} \text{ --- (i)}$$

$$\frac{\partial L}{\partial y} = P_y - \lambda x = 0 \Rightarrow \lambda = \frac{P_y}{x} \text{ --- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = \bar{U}^2 - xy = 0 \text{ --- (iii)}$$

From eqn (i) and (ii) $\Rightarrow$

$$\frac{P_x}{y} = \frac{P_y}{x}$$

$$\Rightarrow y = \frac{P_x x}{P_y} \text{ --- (iv)}$$

Substituting (iv) into utility constraint,

$$x \cdot \frac{P_x x}{P_y} = \bar{U}^2$$

$$\Rightarrow x^2 = \frac{\bar{U}^2 P_y}{P_x}$$

$$\therefore x^* = \frac{\bar{U}^2 \sqrt{P_y}}{\sqrt{P_x}}$$

Now, substituting the value of x into equn (iv).

$$y = \frac{P_x}{P_y} \cdot \bar{U} \sqrt{\frac{P_y}{P_x}}$$

$$\therefore y^* = \bar{U} \sqrt{\frac{P_x}{P_y}}$$

The Hicksian demands are —

$$x^* = \bar{U} \sqrt{\frac{P_y}{P_x}} \quad \text{and} \quad y^* = \bar{U} \sqrt{\frac{P_x}{P_y}}$$

The Expenditure function,

$$\begin{aligned} E(P_x, P_y, \bar{U}) &= P_x \frac{\bar{U} \sqrt{P_y}}{\sqrt{P_x}} + P_y \cdot \frac{\bar{U} \sqrt{P_x}}{\sqrt{P_y}} \\ &= 2\bar{U} \sqrt{P_x P_y} \end{aligned}$$

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$$k_x = \frac{\partial E}{\partial P_x} = ?$$

$$k_y = \frac{\partial E}{\partial P_y} = ?$$

③ $U = \min(x, 2y)$

For this utility function, the goods are perfect complements.
So, consumers will always choose x and y in the ratio
 $x = 2y$.

To achieve a utility level of $\bar{U}$, we need,

$$x^* = \bar{U} \text{ and } y^* = \frac{\bar{U}}{2}$$

These are the Hicksian demands.

Now the expenditure function,

$$\begin{aligned} E(P_x, P_y, \bar{U}) &= P_x x + P_y y = P_x \bar{U} + P_y \frac{\bar{U}}{2} \\ &= \bar{U} \left(P_x + \frac{P_y}{2} \right) \end{aligned}$$

$$h_x = \frac{\partial e}{\partial P_x} = \bar{U}$$

$$h_y = \frac{\partial e}{\partial P_y} = \frac{\bar{U}}{2}$$